

MTL106: Probability and Stochastic Processes
Lecture 11
January 27, 2026

In many experiments an observation is expressible, not as a single numerical quantity, but as a family of several separate numerical quantities. For instance, to record the height and weight of every person in a certain community, we need a pair (x, y) , where the components represent, respectively, the height and weight of a particular person. Similarly, both voltage (x) and current (y) might be of interest in certain electrical experiment. Such experiments can be studied using *multidimensional random variables*.

n-Dimensional Random Variable: The collection $\mathbf{X} = (X_1, X_2, \dots, X_n)$ defined on $(\Omega, \mathcal{F}, P)$ into $\mathbb{R}_n$ by

$$\mathbf{X}(\omega) = (X_1(\omega), X_2(\omega), \dots, X_n(\omega)), \quad \omega \in \Omega,$$

is known as a *n-dimensional random variable* if the inverse image of every *n*-dimensional interval

$$I = \{(x_1, x_2, \dots, x_n) : -\infty < x_i < a_i, \quad a_i \in \mathbb{R}, \quad i = 1, 2, \dots, n\}$$

is also in $\mathcal{F}$, i.e., if

$$\mathbf{X}^{-1}(I) = \{\omega : X_1(\omega) \leq a_1, \dots, X_n(\omega) \leq a_n\} \in \mathcal{F} \text{ for } a_i \in \mathbb{R}.$$

Theorem 1. Let $X_1, X_2, \dots, X_n$ be *n* random variables on $(\Omega, \mathcal{F}, P)$. Then $\mathbf{X} = (X_1, X_2, \dots, X_n)$ is a *n-dimensional random variable* on $(\Omega, \mathcal{F}, P)$.

Proof. Let $I = \{(x_1, x_2, \dots, x_n) : -\infty < x_i < a_i, \quad a_i \in \mathbb{R}, \quad i = 1, 2, \dots, n\}$. Then

$$\begin{aligned} \{(X_1, X_2, \dots, X_n) \in I\} &= \{\omega : X_1(\omega) \leq a_1, X_2(\omega) \leq a_2, \dots, X_n(\omega) \leq a_n\} \\ &= \bigcap_{k=1}^n \{\omega : X_k(\omega) \leq a_k\} \in \mathcal{F}, \end{aligned}$$

as asserted.

CDF of a Two-Dimensional Random Variable: The function $F(., .)$, defined by

$$F(x, y) = P\{X \leq x, Y \leq y\}, \quad \forall (x, y) \in \mathbb{R}_2, \quad (1)$$

is known as the *CDF of the random variable* (X, Y) .

Properties: The two-dimensional CDF has following properties:

1. $F(x, y)$ is nondecreasing and continuous from the right with respect to each coordinate, and
2.
$$\lim_{x \rightarrow +\infty, y \rightarrow +\infty} F(x, y) = F(+\infty, +\infty) = 1,$$

$$\lim_{y \rightarrow -\infty} F(x, y) = F(x, -\infty) = 0, \quad \forall x,$$

$$\lim_{x \rightarrow -\infty} F(x, y) = F(-\infty, y) = 0, \quad \forall y.$$

But the above conditions are not sufficient conditions to make any function $F(.,.)$ a CDF.

Note: Let $x_1 < x_2$ and $y_1 < y_2$. We have

$$\begin{aligned} P\{x_1 < X \leq x_2, y_1 < Y \leq y_2\} &= P\{X \leq x_2, Y \leq y_2\} + P\{X \leq x_1, Y \leq y_1\} \\ &\quad - P\{X \leq x_1, Y \leq y_2\} - P\{X \leq x_2, Y \leq y_1\} \\ &= F(x_2, y_2) + F(x_1, y_1) - F(x_1, y_2) - F(x_2, y_1) \\ &\geq 0 \end{aligned}$$

for all pairs $(x_1, y_1), (x_2, y_2)$ with $x_1 < x_2, y_1 < y_2$.

Theorem 2. A function F of two variables is a CDF of some two-dimensional random variable if and only if it satisfies the following conditions:

1. F is nondecreasing and right continuous with respect to both arguments.
2. $F(-\infty, y) = F(x, -\infty) = 0$, and $F(+\infty, +\infty) = 1$.
3. For every $(x_1, y_1), (x_2, y_2)$ with $x_1 < x_2, y_1 < y_2$, the inequality

$$F(x_2, y_2) - F(x_2, y_1) + F(x_1, y_1) - F(x_1, y_2) \geq 0 \quad (2)$$

holds.

Bivariate Discrete Random Variable: A two-dimensional (or bivariate) random variable (X, Y) is said to be of the discrete type if it takes on pairs of values belonging to a countable set of pairs A with probability 1. We call every pair (x_i, y_j) that is assumed with positive probability p_{ij} a jump point of the CDF of (X, Y) , and call p_{ij} the jump at (x_i, y_j) . Here A is the support of the distribution of (X, Y) .

Clearly, $\sum_{ij} p_{ij} = 1$. As for the CDF of (X, Y) , we have

$$F(x, y) = \sum_B p_{ij},$$

where $B = \{(i, j) : x_i \leq x, y_j \leq y\}$.

Joint Probability Mass Function: Let (X, Y) be a random variable of the discrete type that takes on pairs of values (x_i, y_j) , $i = 1, 2, \dots$ and $j = 1, 2, \dots$. We call

$$p_{ij} = P\{X = x_i, Y = y_j\}, \quad i = 1, 2, \dots, \quad j = 1, 2, \dots,$$

the *joint probability mass function* of (X, Y) .

Theorem 4. A collection of nonnegative numbers $\{p_{ij} : i = 1, 2, \dots; j = 1, 2, \dots\}$ satisfying $\sum_{i,j=1}^{\infty} p_{ij} = 1$ is the PMF of some random variable.